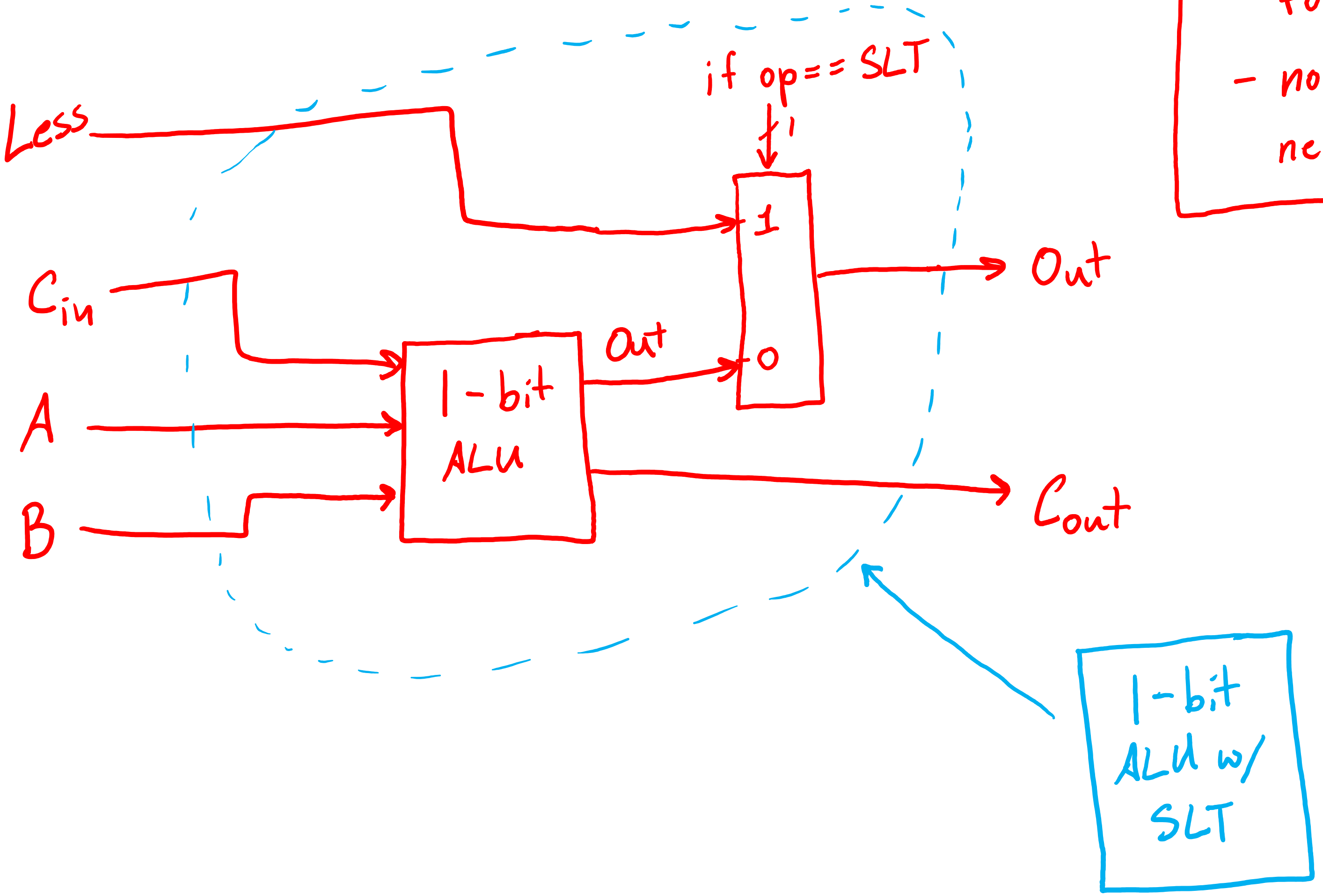
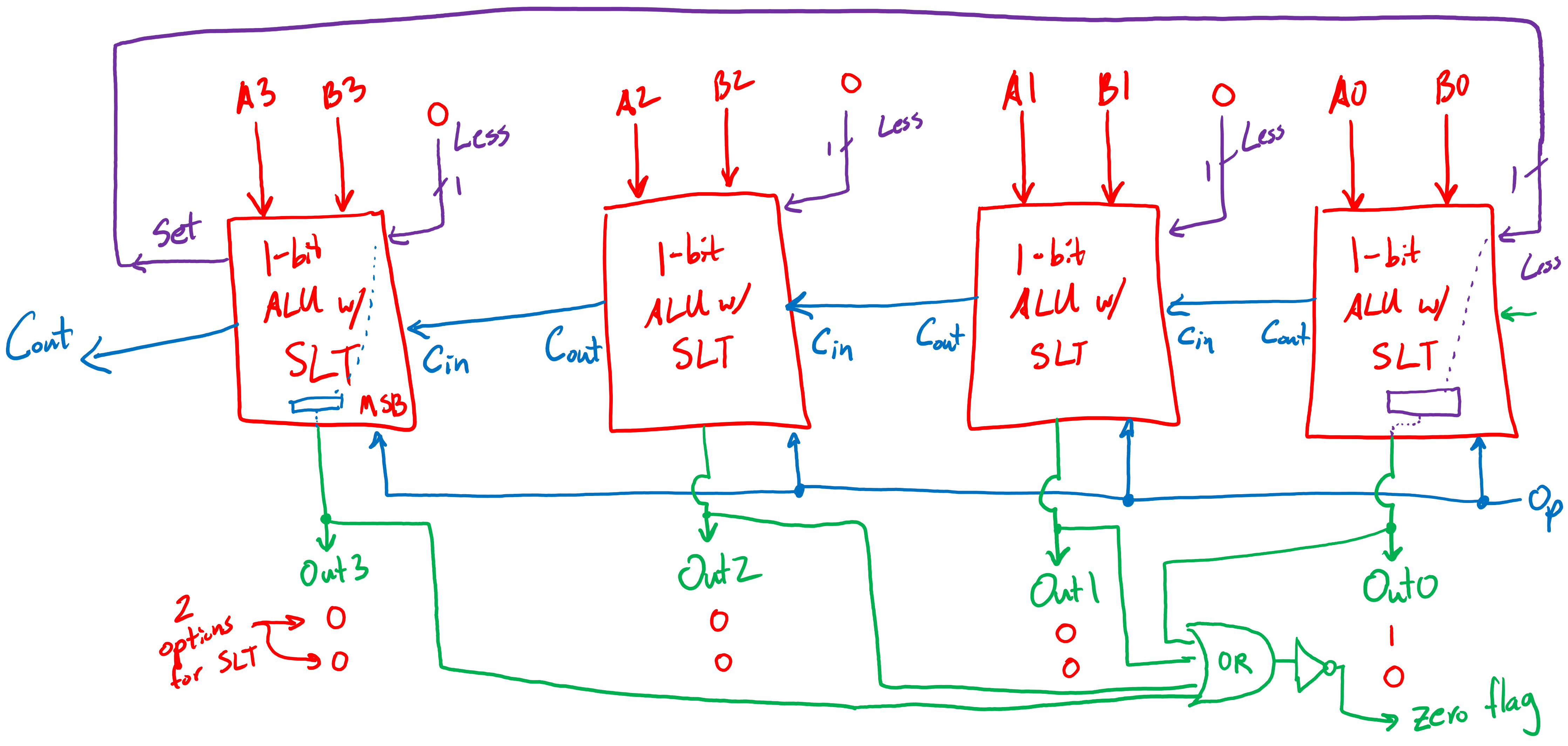
Hardware for Set Less Than

SLT → if (A < B) then Out = 1, else Out = 0



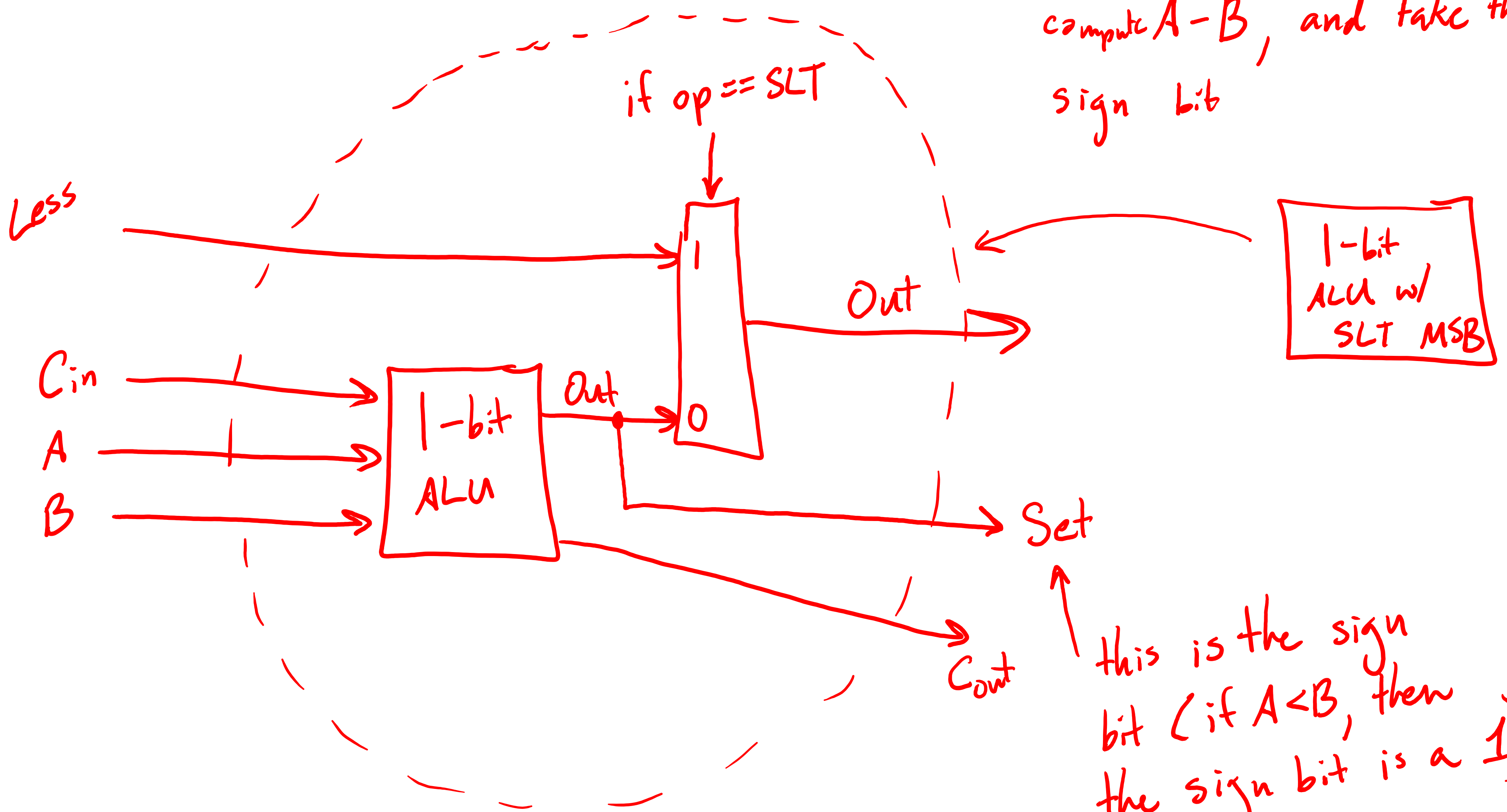
- Homework #1 due next Monday
- Lab 2 progress check today
- no class this Friday or next Monday

Hardware for Set Less Than



Hardware for Most Significant Bit

- have the ALU subtract compute A - B, and take the sign bit



this is the sign bit (if A<B, then the sign bit is a 1)

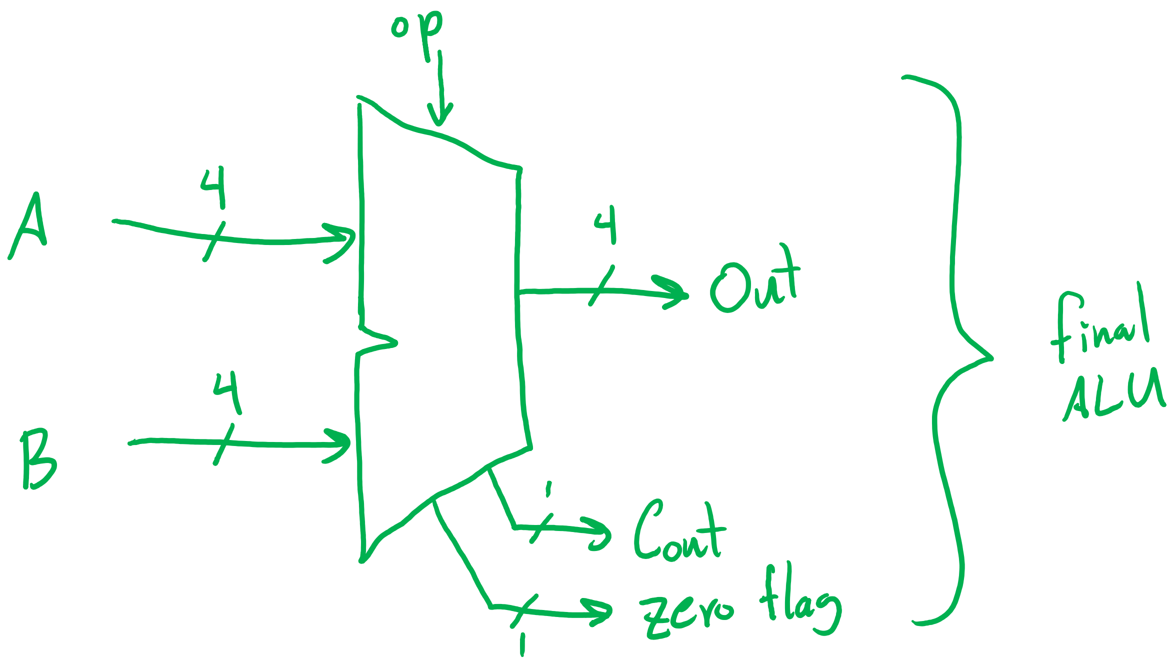
Hardware for Set Less Than

- have the ALUs do a subtract internally

- run the sign bit into the Less input of LSB ALU

Hardware for BNE and BEQ

- zero flag is set high only when the ALU output is the number 0



Datapath and Control

4.1-4.4

Goal

- Build an architecture to support the following instructions:

arithmetic: add, sub, addi, slt

memory: lw, sw

branches: beq, j

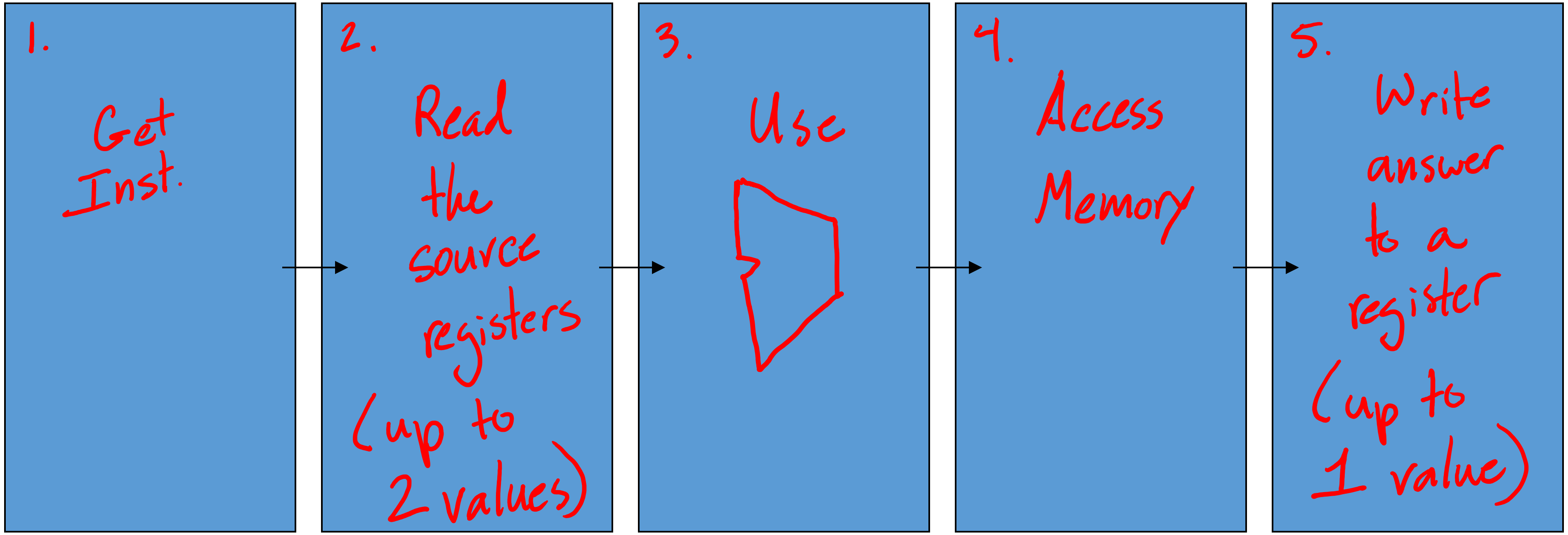
MIPS Instruction Trends

- Read 2 regs: add, sub, slt, beq, sw
  3 regs: X
  1 reg: addi, lw
  0 reg: j
- Write: add, sub, addi, lw, slt

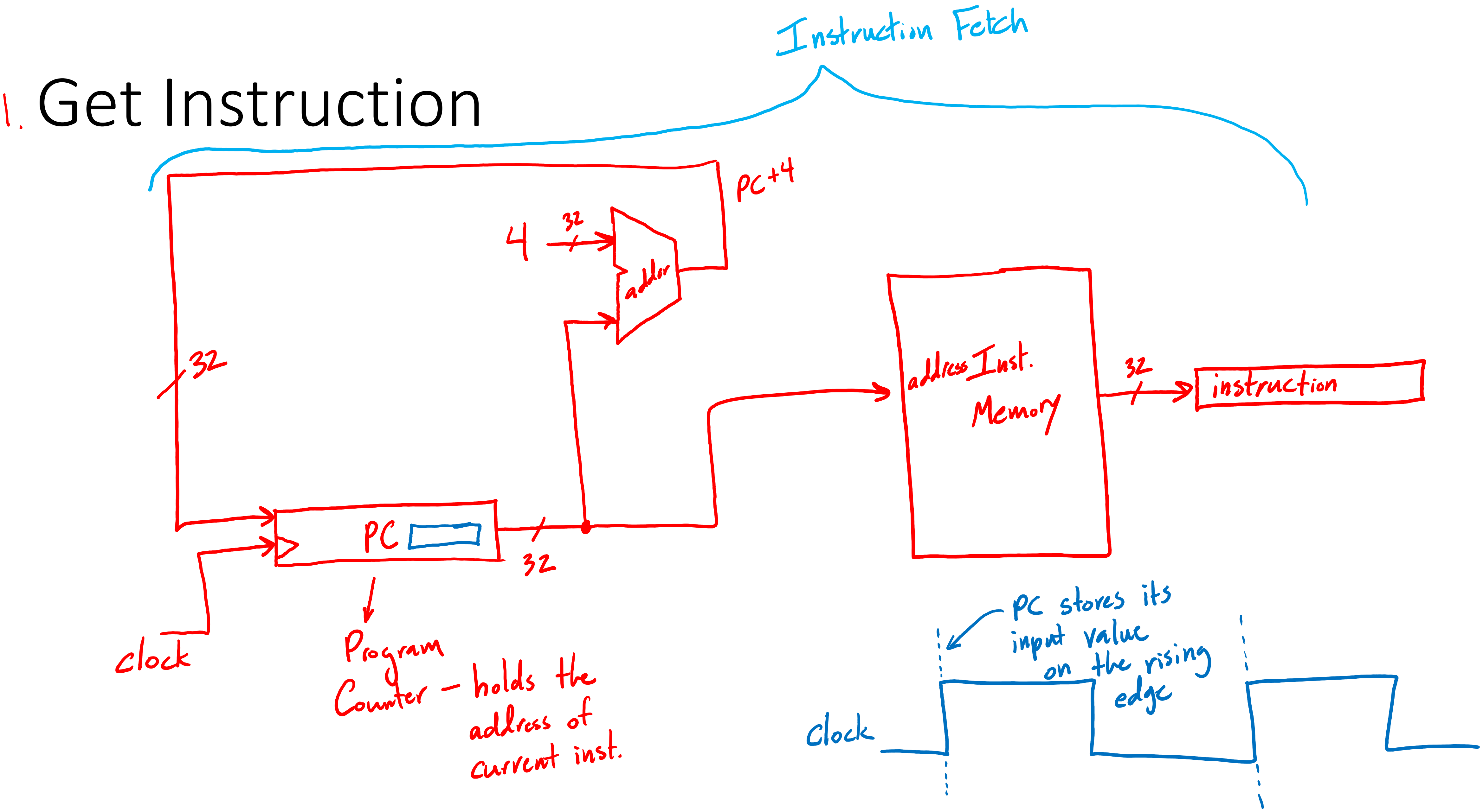
Framework

- Makefile
  - make lab2
- hand in all java source and the Makefile

- Lab 2
  - name top level class 'lab2'



1. Get Instruction



Instruction Fetch

- Key Points:
  - PC is a register (clocked unit)
  - updates on the rising edge
  - inst. memory is not clocked unit

2. Register File – structure that holds all the register data

- 32 regs. x 32 bits
- Read 2 registers
- Write 1 register

Register File